

GANESH RUN

THE GREAT LADDU CHASE — COMPLETE TOOLS & TECHNICAL SPECIFICATION

Developer: Manikanta Gara (NIAT x NSRIT) **3D Engine:** Three.js (WebGL 2.0) **Performance:** 60 FPS Locked

Production URL: <https://ganesh-runner.vercel.app>

🔧 1. Core Frameworks, Runtimes & Build System

Vite Bundler & Dev Server

V6.4.3

Modern frontend build tool utilizing Rollup under the hood. Provides instant Hot Module Replacement (HMR) and compiles production assets into a minified, high-performance static distribution bundle.

Node.js Runtime

V20.X / V24.X

Underlying execution engine for build pipelines, continuous integration, headless deployment orchestration, and mathematical invariant simulation testing (32,400m test runs).

📦 2. Production Libraries & Dependencies

Library	Version	Category	Role in Ganesh Run
Three.js	^0.186.0	3D WebGL Graphics Engine	Renders the entire 3-track railway scene, 3D trains, animated characters, dynamic shadows, volumetric Xenon headlight beams, and tracking camera.
canvas-confetti	^1.9.4	Particle FX System	High-performance GPU-accelerated confetti celebration triggered on high scores, level milestones, and game over screens.

🌐 3. Native Web Platform APIs (Zero-Dependency Engine)

WebGL 2.0

GPU

Hardware-accelerated GPU pipeline delivering locked 60 FPS graphics across smartphones, tablets, and desktop browsers.

Web Audio API

SYNTH

Procedural sound synthesis: Raga Bhoopali melodies, Tanpura drones, Dholak percussion, and dual-tone train horns without external MP3 files.

Touch Gestures

MOBILE

Low-latency swipe gesture detector capturing touchstart, touchmove, and touchend vectors for dodging, jumping, and sliding.

HTML5 Canvas 2D

2D CONTEXT

Powers procedural texture synthesis in RealisticCharacterTextures.js, creating ballast gravel, WAP-7 liveries, and hazard stripes.

Web Storage

LOCALSTORAGE

Persists lifetime high scores, distance records, total laddus, and selected Avatar preference across browser sessions.

Import Maps

W3C STANDARD

Native module resolution for Three.js, guaranteeing fail-safe execution even if raw unbundled source files are loaded.

 4. Graphical & Visual Assets Inventory ([public/assets/](#))

Asset Filename	Size	Format	Role in Ganesh Run
bal_ganesha_preview.png	843.7 KB	PNG (Alpha)	Preview photo card for Bal Ganesha in the Avatars modal.
ganesha_front_run.png	843.7 KB	PNG (Alpha)	Front-facing 3D render of Lord Ganesha for Main Menu & Pause screens.
ganesha_back_run.png	898.7 KB	PNG (Alpha)	Dynamic rear running sprite of Bal Ganesha for active track gameplay.
mushika_preview.png	185.0 KB	PNG (Alpha)	3/4 isometric preview photo for the "Ganesha on Mushika" avatar card.
mushika_front.png	176.1 KB	PNG (Alpha)	Front 3D render of Ganesha seated on Mount Mushika granting abhaya mudra.
mushika_back.png	147.9 KB	PNG (Alpha)	Rear dynamic running sprite of Ganesha riding Mushika on the track.
mushika_vahana_sheet.png	792.8 KB	PNG (Alpha)	Multi-angle stride texture sheet for Mount Mushika's galloping animation.
shiva_back_run.png	792.4 KB	PNG (Alpha)	High-res running sprite of Lord Shiva holding Trishul chasing Ganesha.
ref_front.jpgg & ref_back.jpgg	1,014 KB	JPEG	Master source photorealistic character reference renders.

 5. Procedural 3D Models & Corridor Engine

Indian Railways WAP-7 Locomotive PROCEDURAL 3D

8.8m electric locomotive with authentic Indian Railways blue livery, chevron hazard striping, pantographs, bogies, cast steel wheels, and Xenon headlight beams.

ICF / LHB Passenger Coaches PROCEDURAL 3D

9.2m passenger coaches with corrugated steel roofs, double bogies, rubber gangway bellows, and rear glowing red safety tail-lights.

Auto-Rickshaws & Tempos CITY OBSTACLES

Vivid Kelly green & sunshine yellow auto-rickshaws (tuk-tuks) and delivery tempos with safety bumpers and rooftop taxi indicator lights.

Guaranteed Open Track Corridors CLEARANCE RULE

Strict mathematical invariant: maximum 2 stationary trains per wave, leaving exactly 1 track 100% open with a trail of 6 glowing Motichoor Laddus.

 6. Typography & UI Design Tokens

Font Family	Weights	Provider	Role
Cinzel Decorative	700, 900	Google Fonts	Game title ("GANESH RUN"), modal headers, victory badges.
Outfit	400, 600, 700, 800	Google Fonts	Modern UI: buttons, guide bar, settings, HUD score readouts.
JetBrains Mono	400, 600	System Monospace	Score meters, distance counters, and calibrated gyro angles.

 7. DevOps & Cloud Hosting Infrastructure

Vercel Edge Network PRIMARY PRODUCTION

Hosts the Vite compiled static build at <https://ganesh-runner.vercel.app> with global CDN routing and automatic Gzip compression.

GitHub Pages & REST Deployer MIRROR & CI/CD

Automated zero-git push script syncing to **MANIKANTA-GARA/Ganesh-Runner** with live mirror at <https://MANIKANTA-GARA.github.io/Ganesh-Runner/>.